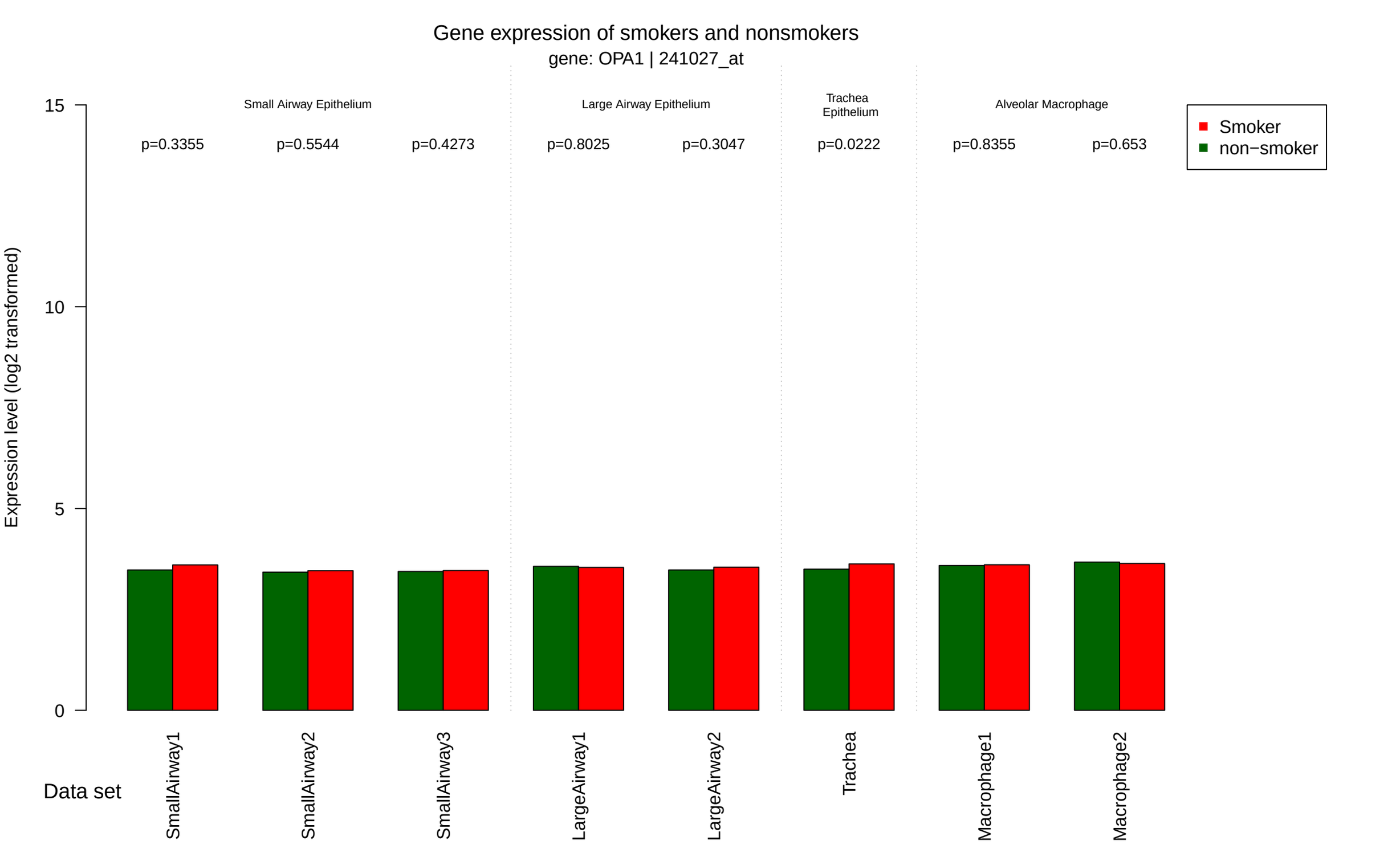
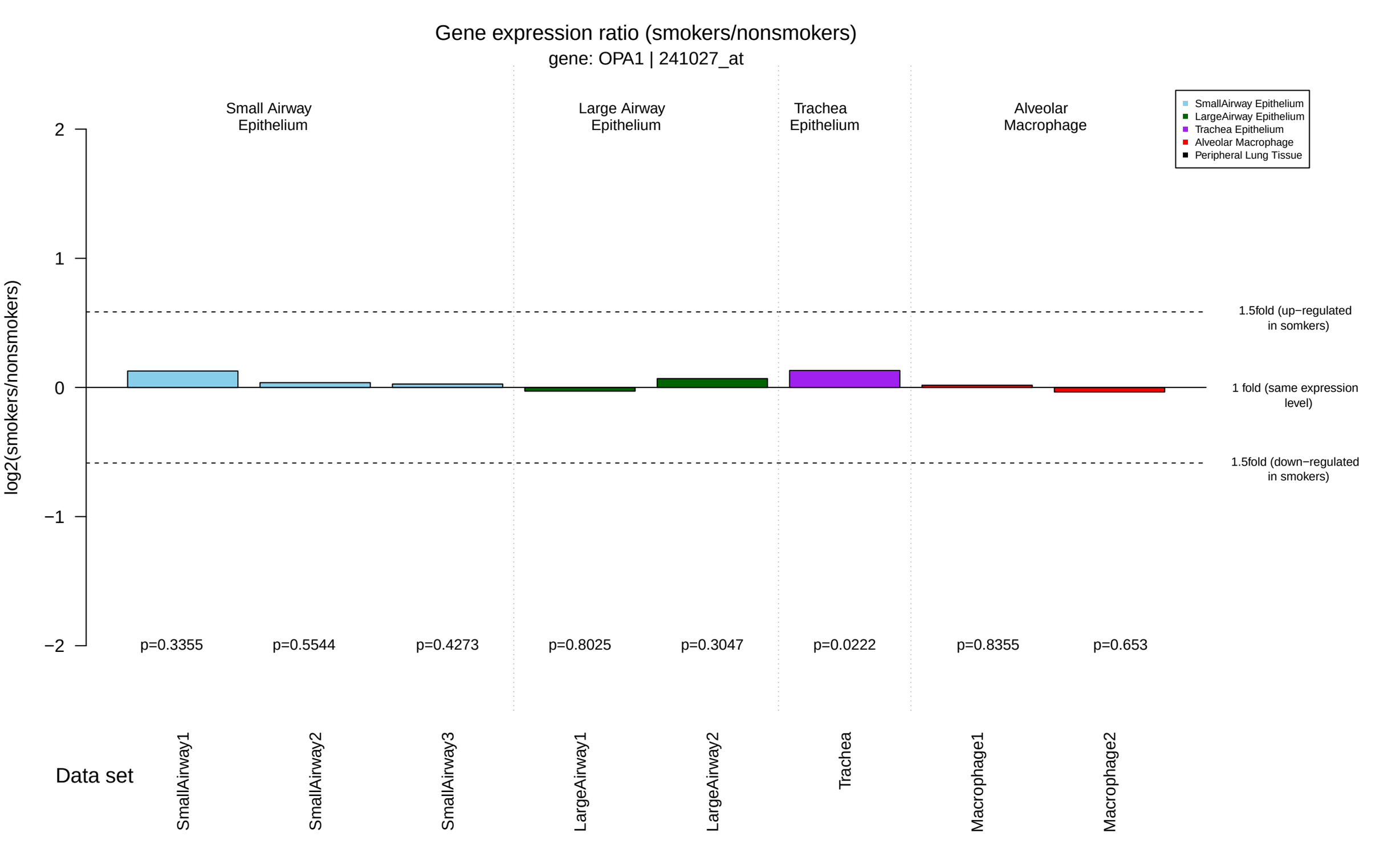
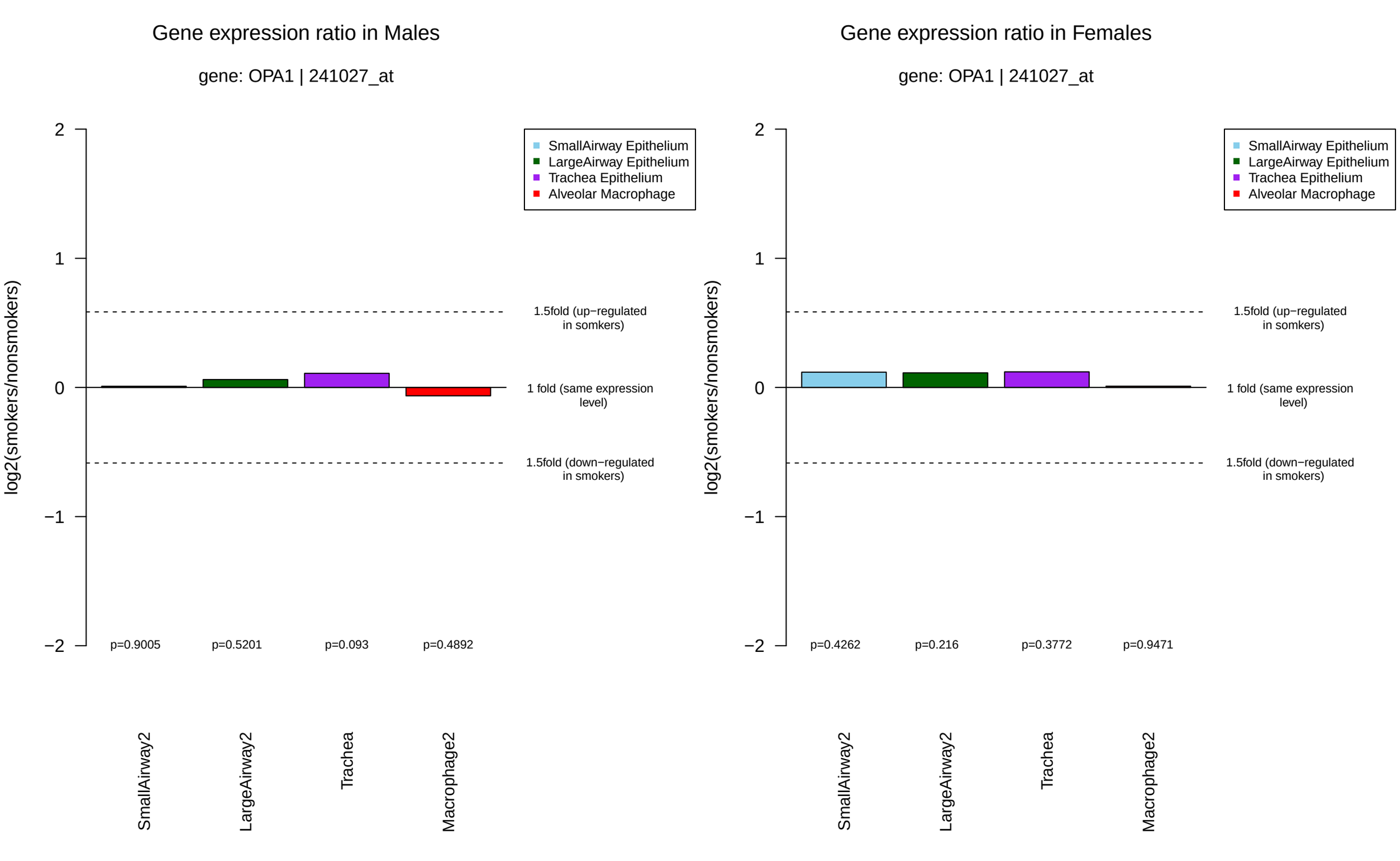
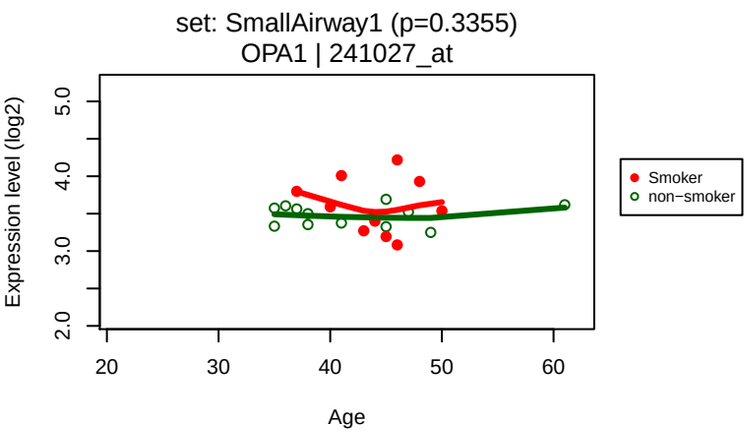
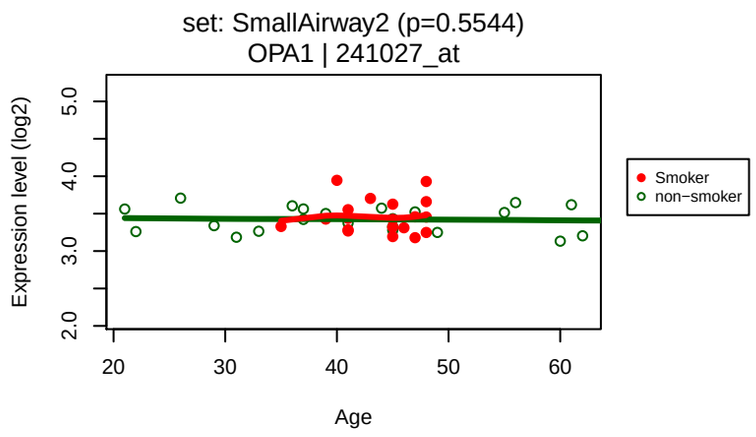
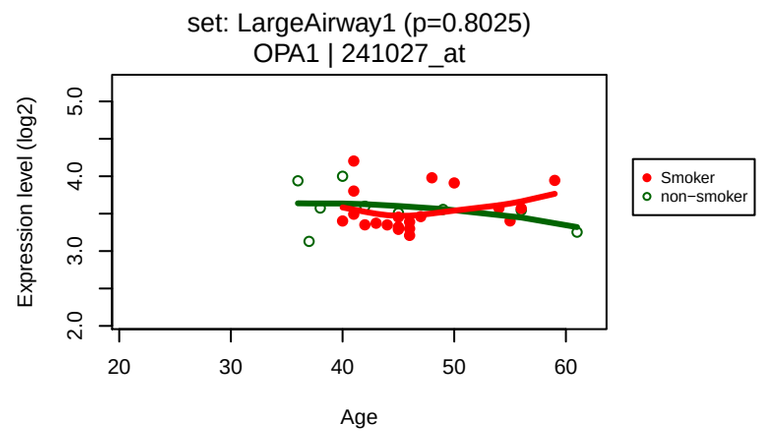

Gene expression of smokers and nonsmokers
gene: OPA1 | 241027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3355
p=0.5544
p=0.4273
p=0.8025
p=0.3047
p=0.0222
p=0.8355
p=0.653
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

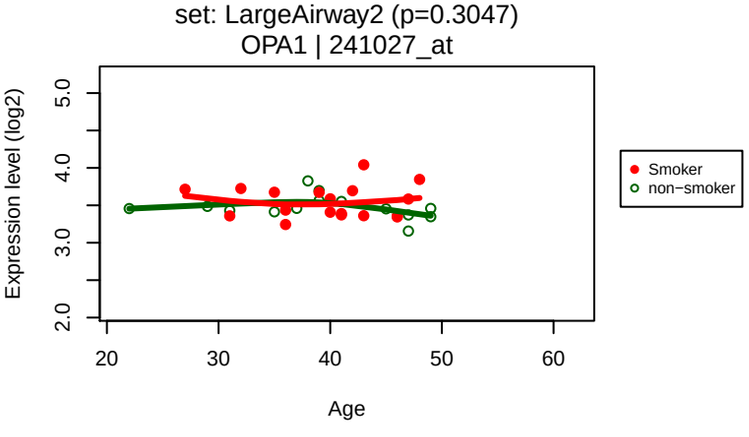

Gene expression ratio (smokers/nonsmokers)
gene: OPA1 | 241027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3355
p=0.5544
p=0.4273
p=0.8025
p=0.3047
p=0.0222
p=0.8355
p=0.653
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

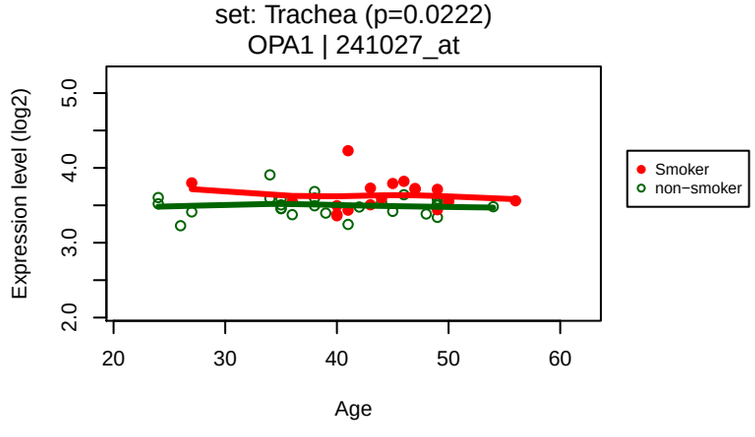

Gene expression ratio in Males
gene: OPA1 | 241027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9005
p=0.5201
p=0.093
p=0.4892
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OPA1 | 241027_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4262
p=0.216
p=0.3772
p=0.9471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

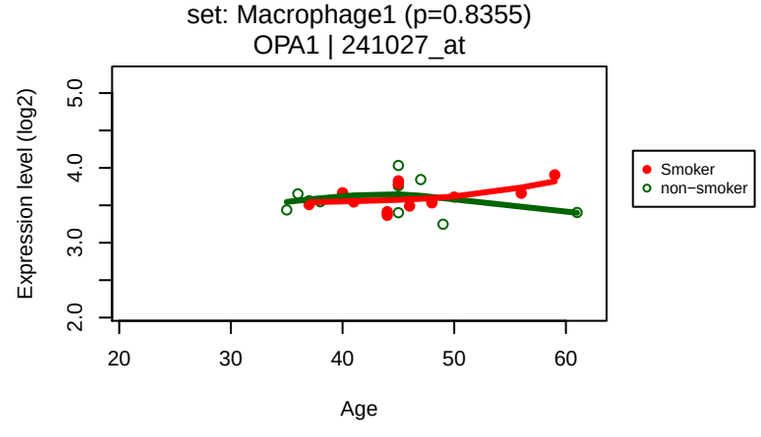

set: SmallAirway1 (p=0.3355)
OPA1 | 241027_at
Expression level (log2)
Age
Smoker
non−smoker

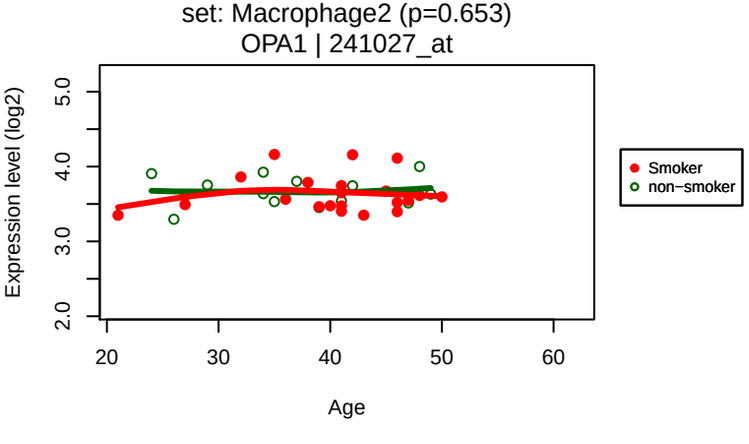

set: SmallAirway2 (p=0.5544)
OPA1 | 241027_at
Expression level (log2)
Age
Smoker
non−smoker

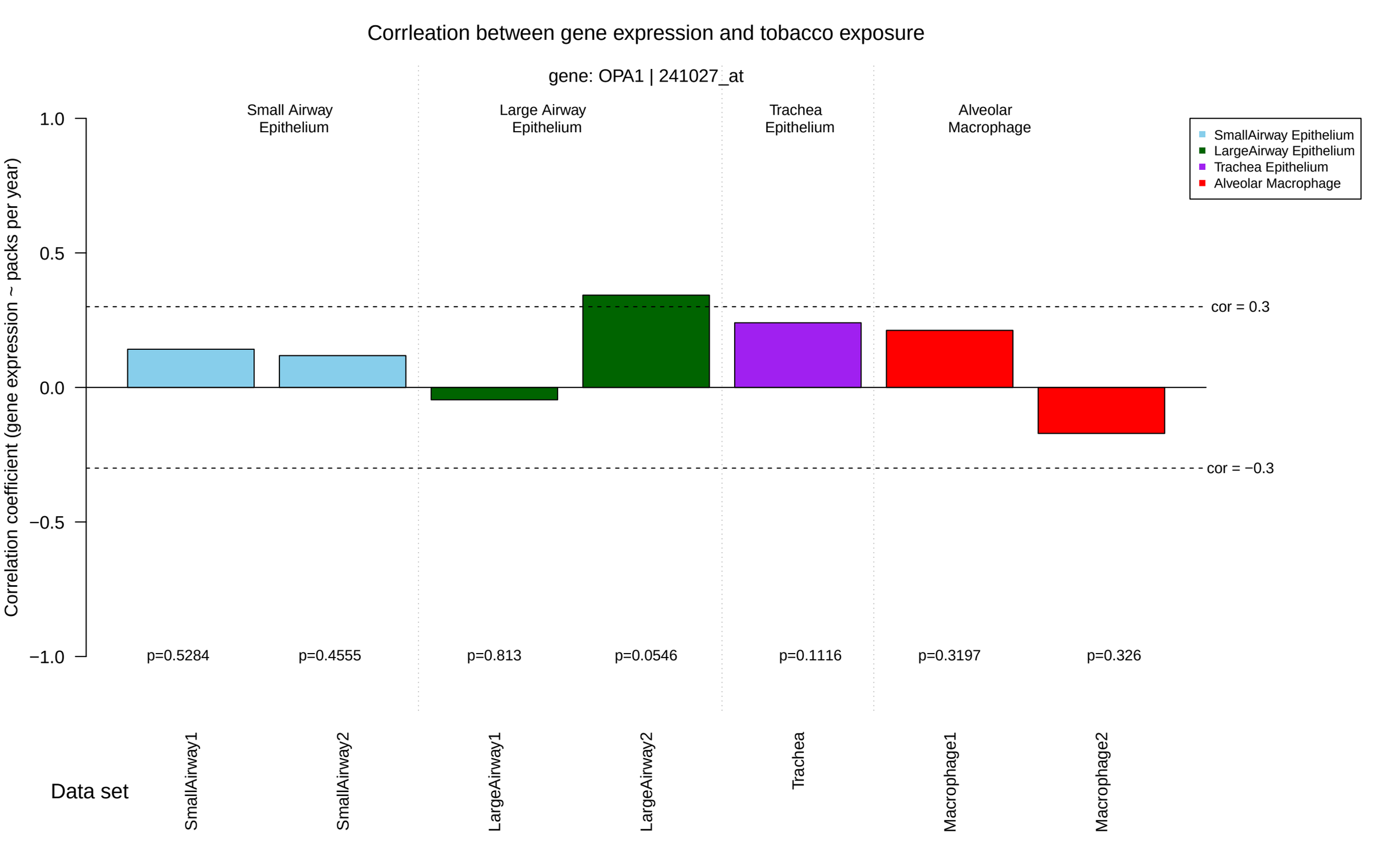

set: LargeAirway1 (p=0.8025)
OPA1 | 241027_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3047)
OPA1 | 241027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0222)
OPA1 | 241027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8355)
OPA1 | 241027_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.653)
OPA1 | 241027_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OPA1 | 241027_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5284
p=0.4555
p=0.813
p=0.0546
p=0.1116
p=0.3197
p=0.326
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2